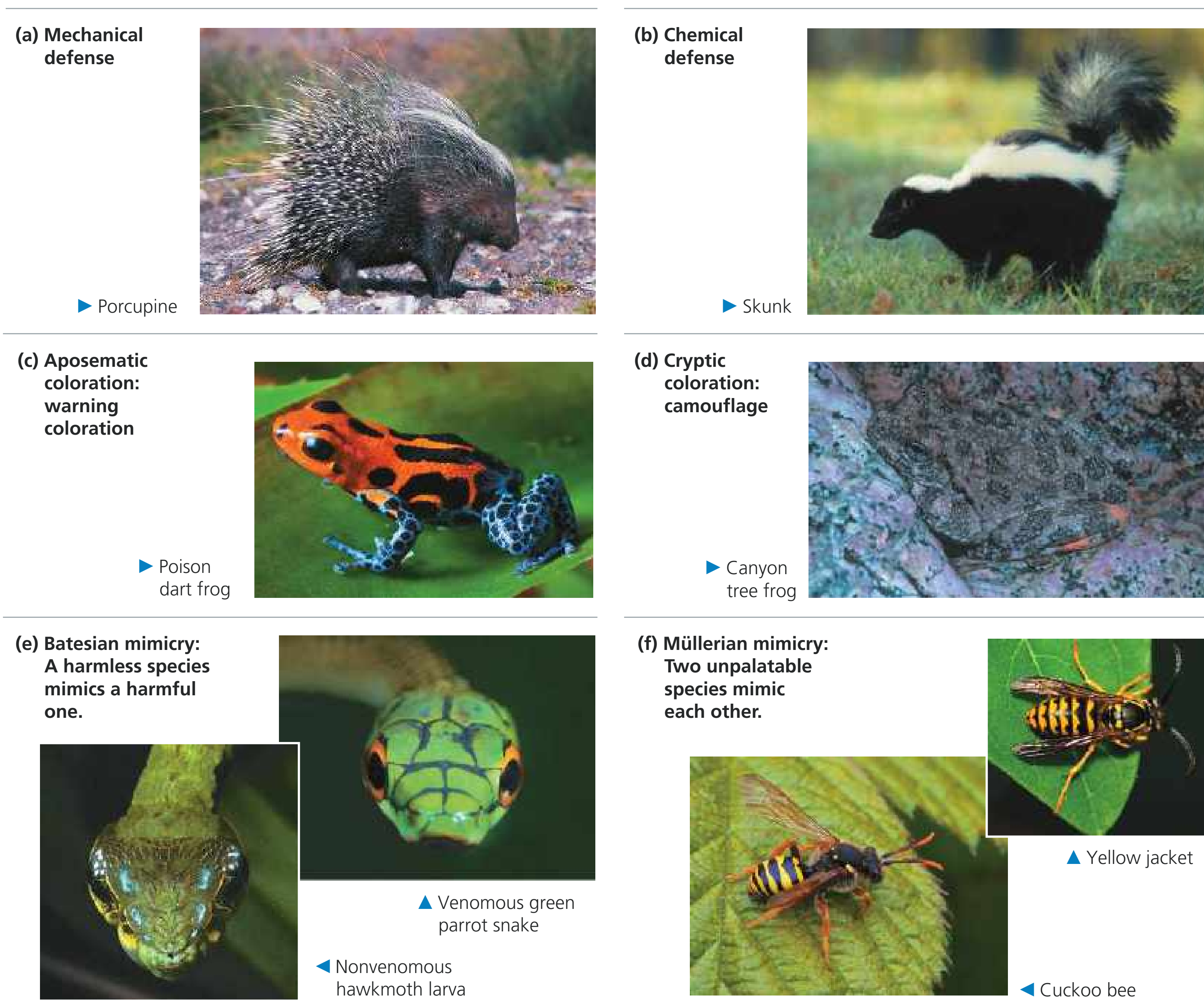
you can interpret data on the impact of natural selection for a specific predator-prey interaction.

Many important feeding adaptations of predators are obvious and familiar. Most predators have acute senses that enable them to find and identify potential prey. Rattlesnakes and other pit vipers, for example, find their prey with a pair of heat-sensing organs located between their eyes and nostrils (see Figure 50.7b). Owls have characteristically large eyes that help them see prey at night. Many predators also have adaptations such as claws, fangs, or poison that help them catch and subdue their food. Predators that pursue their prey are generally fast and agile, whereas those that lie in ambush are often disguised in their environments.

Just as predators possess adaptations for capturing prey, potential prey have adaptations that help them avoid being eaten. In animals, these adaptations include behavioral defenses such as hiding, fleeing, and forming herds or schools. Active self-defense is less common, though some large grazing mammals vigorously defend their young from predators.

Animals also display a variety of morphological and physiological defensive adaptations. Mechanical or chemical defenses protect species such as porcupines and skunks (**Figure 54.6a** and **b)**. Some animals, such as the European fire salamander, can synthesize toxins; others accumulate toxins passively from the plants they eat. Animals with effective chemical defenses often exhibit bright

▼ **Figure 54.6 Examples of defensive adaptations in animals.**



MAKE CONNECTIONS *Explain how natural selection could increase the resemblance of a harmless species to a distantly related harmful species. In addition to selection, what else could account for a harmless species resembling a closely related harmful species? (See Concept 22.2.)*

Mastering Biology HHMI Video: Moth Mimicry: Using Ultrasound to Avoid Bats